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Approach to Peripheral Neuropathies Leading to Hospital Admission

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Peripheral neuropathies present with varying degrees of sensory, motor, and autonomic symptoms, or neuropathic pain.[1] Peripheral neuropathies may prompt hospitalization, often because of the degree of disability or rapid progression relative to neuropathies encountered in the outpatient setting. We propose a stepwise approach to diagnosis in this patient population (Figure).

Evaluation begins with identifying the temporal evolution and characterizing the clinical phenotype. Diagnostic testing is guided by these features and should generally be approached from minimally invasive to more invasive testing as clinically appropriate.

Author Contributions

R.M. Hass: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A.R. Santilli: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J.M. Martinez-Thompson: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M.V. Pinto: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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Reference

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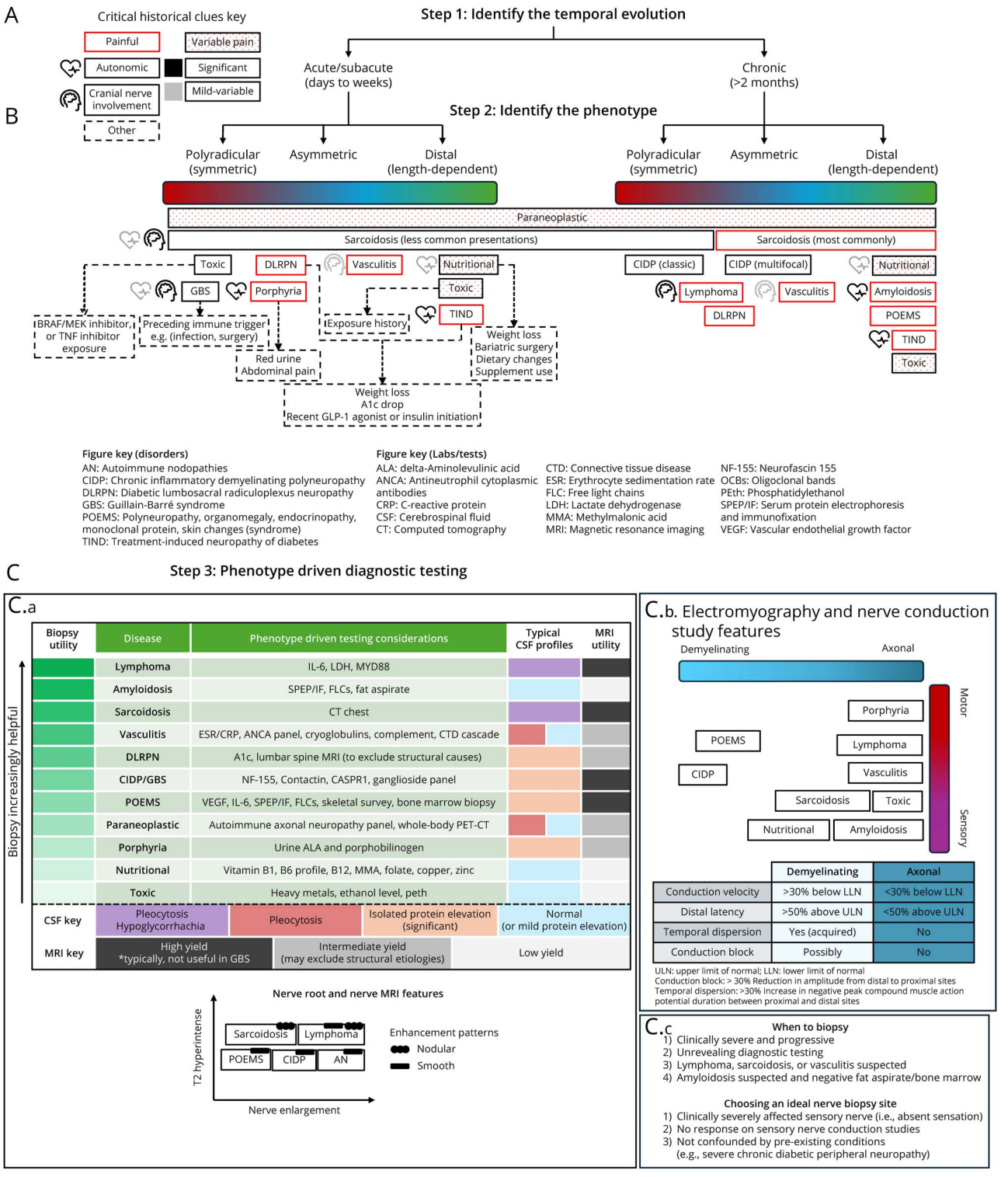
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Figure A Stepwise Approach to Hospital Peripheral Neuropathies



Evaluation starts by identifying the temporal evolution (A. Step 1). Typical phenotypic presentations of peripheral nerve disorders, identified through neuromuscular examination and screening for relevant historical clues, are shown in B. Step 2. From here, an appropriately tailored diagnostic work-up based on suspected etiology can be conducted (Step 3), typically starting with differential driven laboratory testing (C.a Step 3). Electrodiagnostic testing (C.b Step 3) is most useful in identifying suspected demyelination, extending pattern assessment when clinically uncertain, and in choosing nerve biopsy targets (C.c Step 3).

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